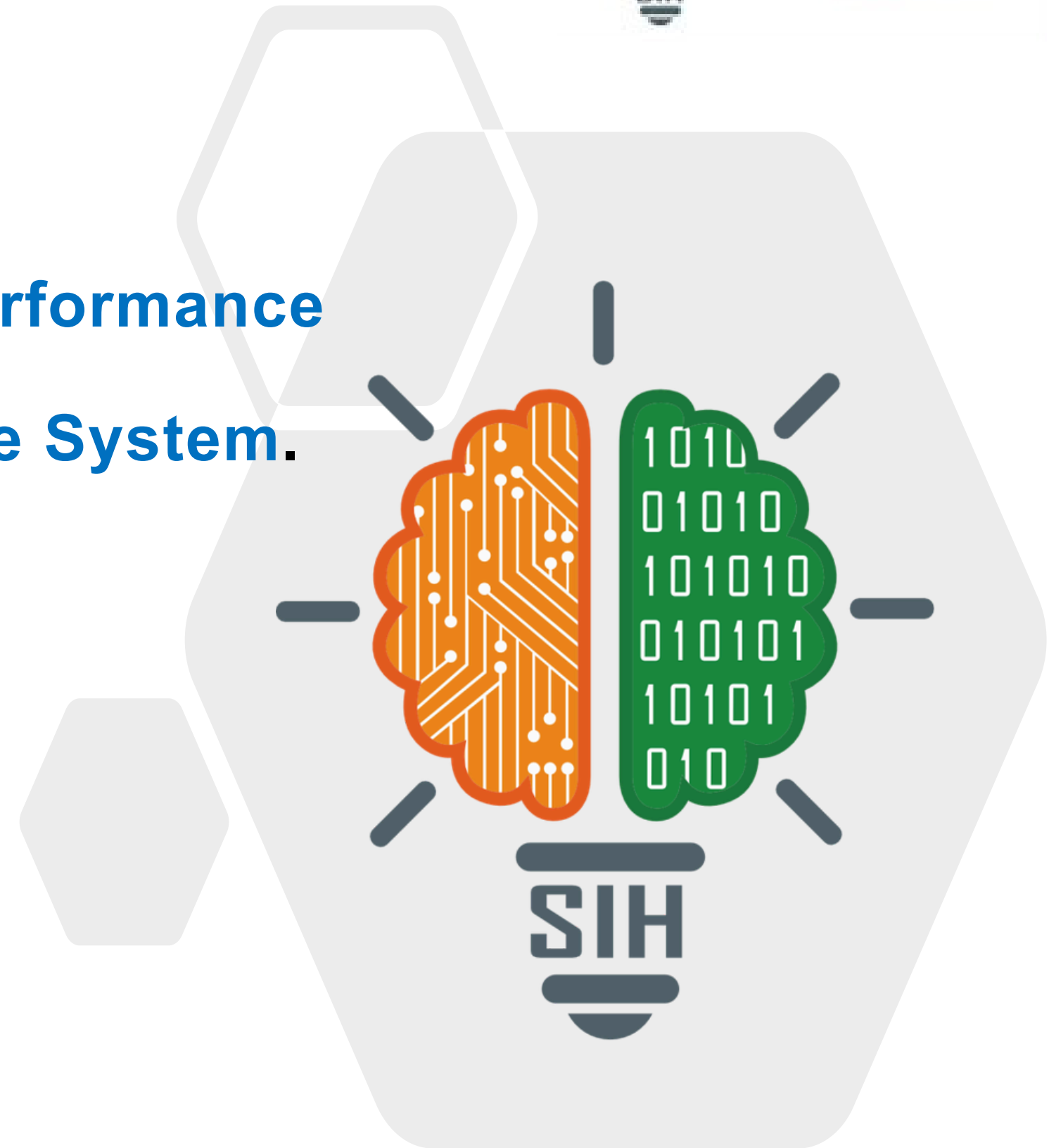


SMART INDIA HACKATHON 2026



- Problem Statement ID – **SIH26050**
- Problem Statement Title- **High Altitude Performance Optimization and Robust Design of Anti-Drone System.**
- Theme- **Robotics and Drones**
- PS Category- **Hardware**
- Team ID-
- Team Name - **Naan-Sense**



SIA-KAVACH

Sense. Adapt. Track. Defend.



PROPOSED SOLUTION MULTI-MODAL SENSING

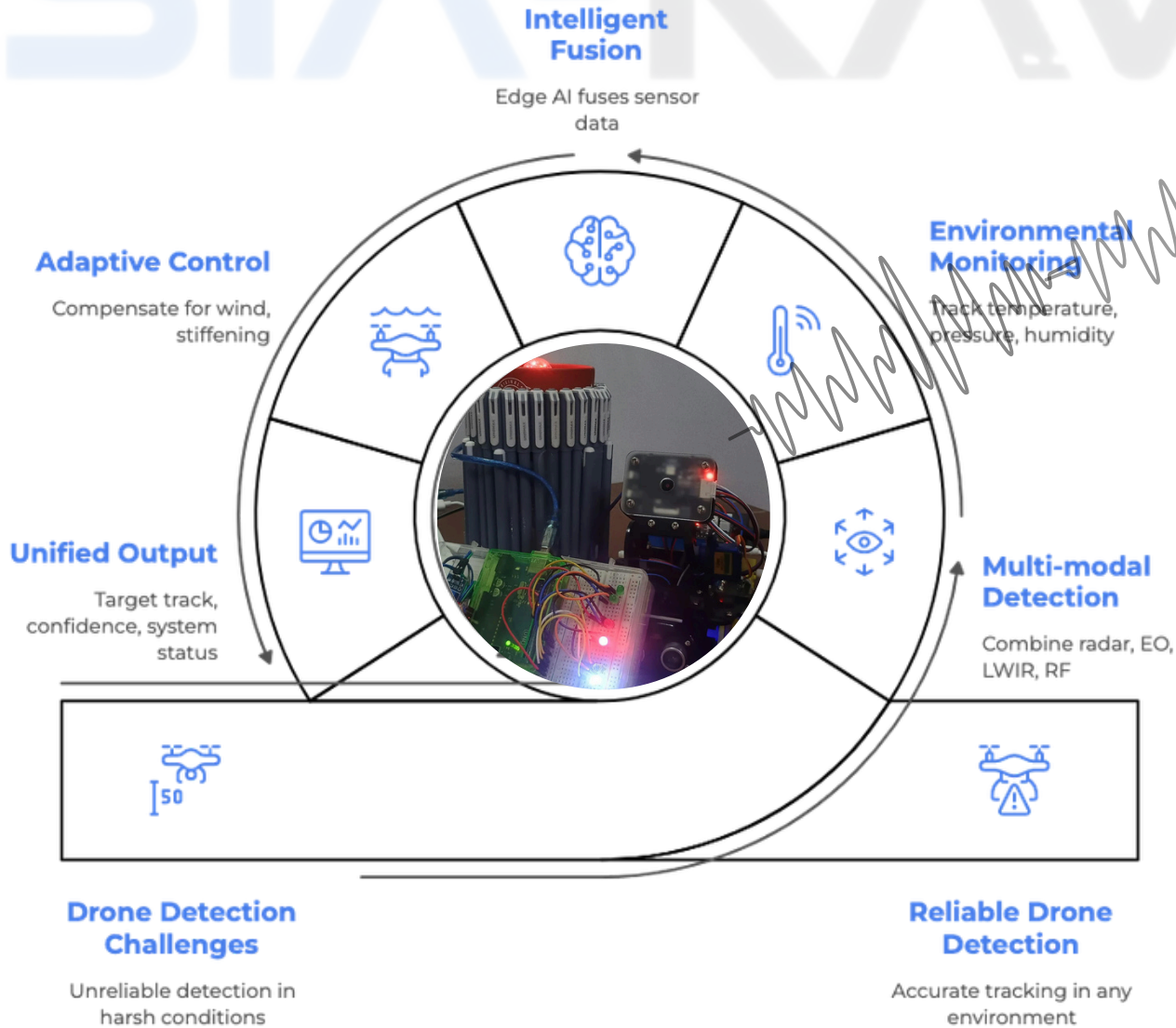
Radar + EO + LWIR + Passive RF
FMCW/mmWave Radar • EO Camera •
LWIR Thermal • Passive RF

System Monitoring
Temperature • Pressure • Humidity •
Vibration • Battery • Actuators

Intelligent Processing
Edge AI • EKF • Weighted
Sensor Fusion

Adaptive Control
Real time compensation for wind,
mechanical stiffening and system
degradation

Unified Output
Target Track + Confidence Score +
System Health



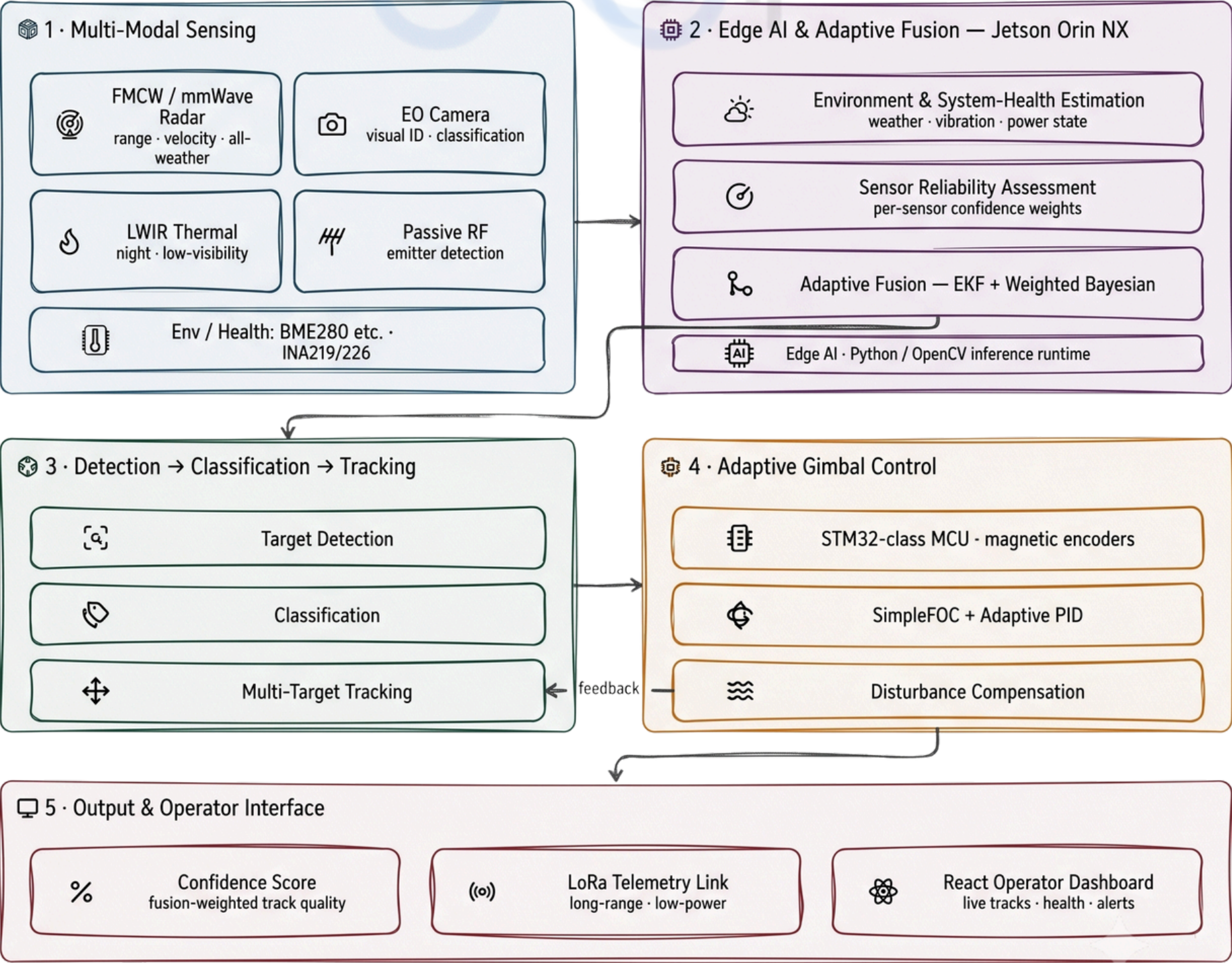
We developed a custom drone prototype to conduct preliminary testing and validate the effectiveness of our Anti-Drone System.

CLOSED LOOP ARCHITECTURE
Sense → Assess → Fuse → Track → Adapt → Correct
It detects the threat and monitors its own ability to detect it.
MULTI SENSORS SMARTER DECISIONS

BUILT TO OVERCOME
Extreme Cold → Thermal & power-aware control
High Winds → Encoder feedback + adaptive PID
Low Pressure → Environmental & thermal zone monitoring
Fog / Snow / Dust → Reduce unreliable EO confidence
Vibration → Isolation + EKF noise correction
Connectivity Loss → Edge processing + long range telemetry

INNOVATION EDGE
Environment Aware: Conditions directly influence system decisions.
Dynamic Fusion: Sensor weights adapt to real time reliability.
Self Diagnosing: Detects degradation in sensors, batteries, actuators and tracking.
Adaptive Tracking: Control parameters automatically compensate for disturbances.
Multi Modal: Radar + RF + EO + LWIR reduces single sensor dependency.

Adaptive Multi-Modal Detection & Tracking Pipeline



Components & Technology



Jetson-Nano Orin



LoRa Module



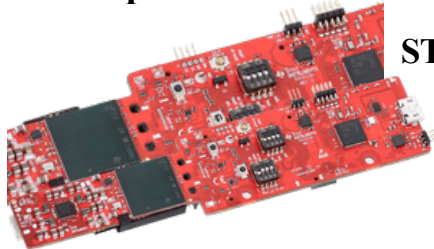
Eo Camera



FLIR Lepton 3.5 LWIR



STM32 MCU

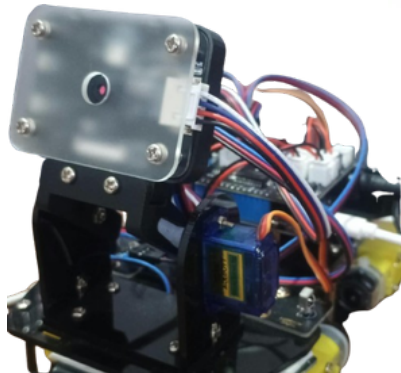
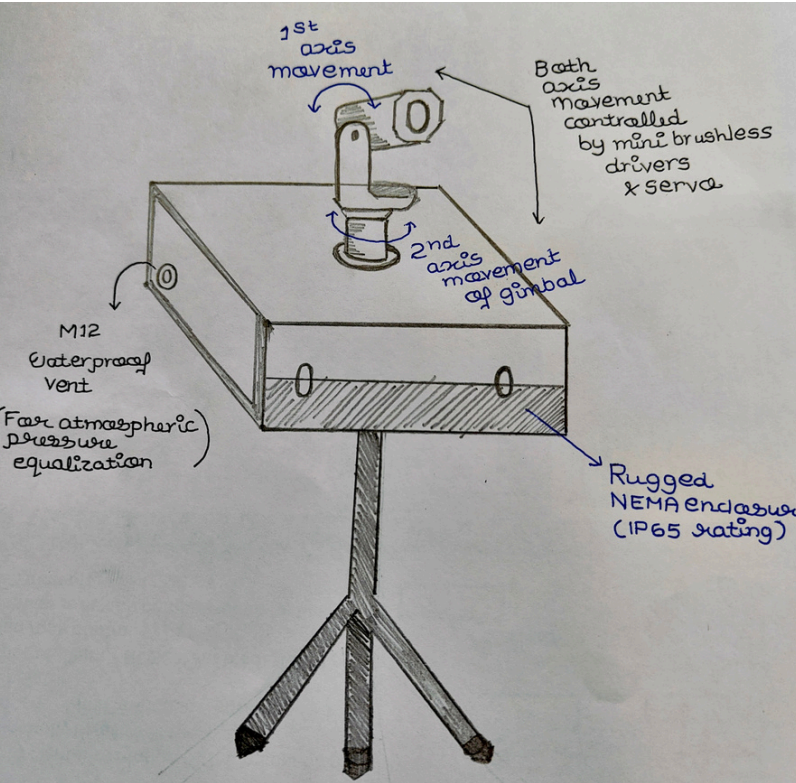


TI IWR6843
mmWave Radar



OpenCV

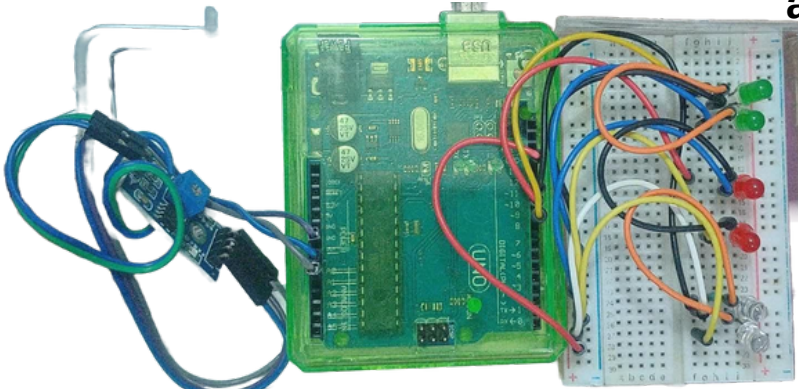
Hardware



Thermal Camera
Gimble

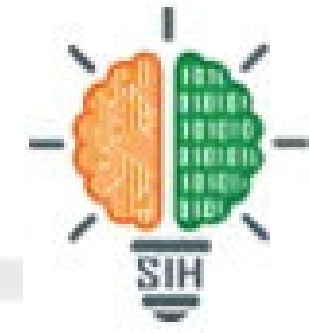


Signalling Unit
and Neutralizer



Main Control Unit

((For basic simulation we are currently using Arduino))



FEASIBILITY

KEY CHALLENGES

MITIGATION STRATEGY



COTS & Modular Design: Uses accessible components for rapid prototyping and easy upgrades.

Climate Extremes: Sub-zero battery drain and low pressure electronics overheating.

Thermal Resilience: LiFePO₄ batteries, smart BMS, and active zoned heating.

Edge AI & LoRa: Local Jetson processing and radio telemetry require zero cloud connectivity.

Mechanical Stress: High-altitude wind gusts and platform vibration throwing off tracking.

Adaptive Stabilization: Encoder feedback loops and Extended Kalman Filter (EKF) noise suppression.

Closed-Loop Feedback: Software is designed to self monitor and correct mechanical lag instantly.

Sensor Blindness: Sudden fog, snow, or dust rendering optical cameras useless.

Dynamic Fusion: AI actively shifts trust away from blinded sensors toward Radar/Thermal.



SODIUM-ION HIGH-ALTITUDE EDGE



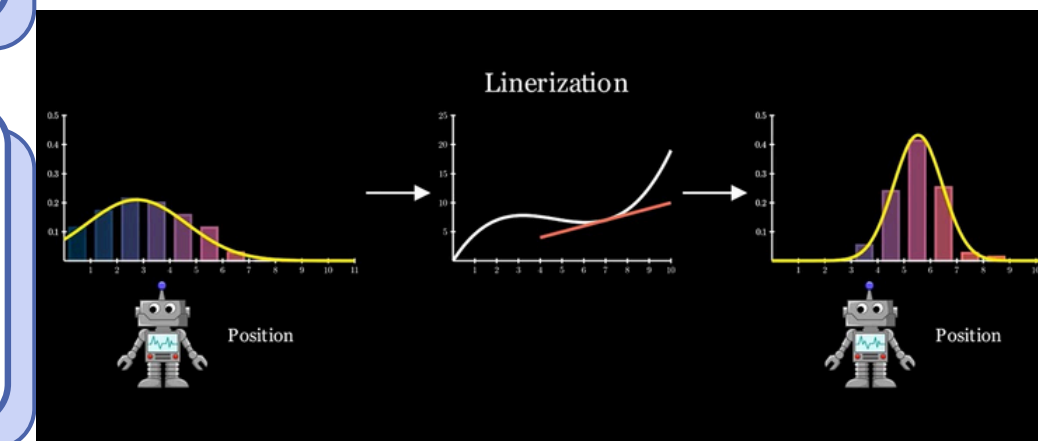
-40°C COLD TOLERANCE
No capacity loss in severe frost



SUPERIOR SAFETY
Zero thermal runaway risk



RUGGED & DURABLE
Resists high-altitude vibration



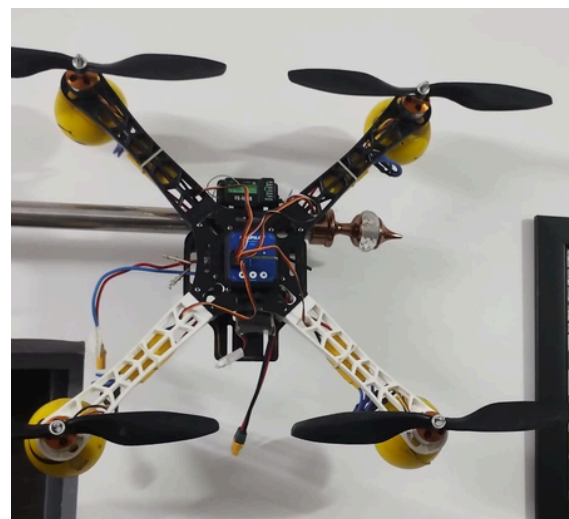
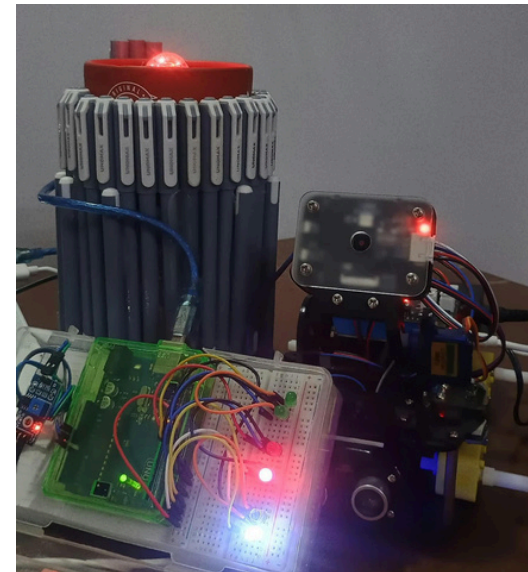
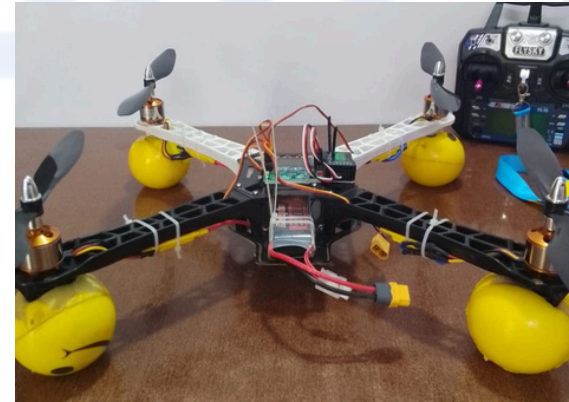
Potential Impact on the Target Audience

Reliable UAV tracking:
even in volatile high altitude conditions.

Real time visibility:
context aware threat assessment

Low latency Edge AI:
no dependency on external networks.

Key Outcome:
Detect - Understand - Adapt - Track



Benefits of Solution

Strategic Resilience:
self-diagnosing, self-correcting & active environmental adaptation
ensuring high tracking stability despite wind and vibration.

Economic Scalability:
simplifies maintenance, easy hardware upgrades, cost-effective
deployment scaling without requiring full system replacements.

Future-Proof Capability:
clear expansion path from single-target interception to multi-
target tracking, swarm tracking, and integration into a distributed
sensor network.

Operational Reliability:
Multi-modal sensor fusion reduces the dependence on single sensor,
ensuring the system continues to function effectively through
changing weather and system conditions.

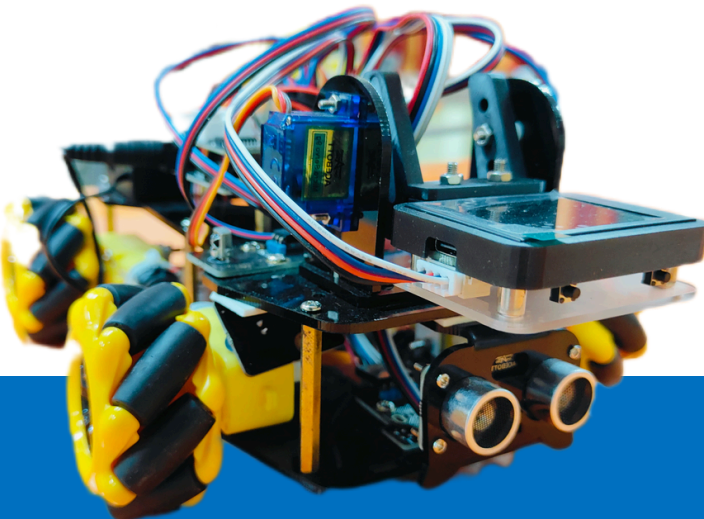
Bibliography

- [DRDO's Radar-Based Drone Jammer with anti drone cope cage](#)
- [Ghatak UCAV Program](#)
- [DRDO Anti Drone System Overview](#)
- [Integrated Drone Detection and Interdiction System \(Mark-2\)](#)

[Research Document](#)
(Clickable)



Demonstration Video



Parameter	Existing DRDO D4 System (Baseline)	Proposed Sia-Kavach (High-Altitude Optimised)
Cost-Effectiveness	High unit cost; designed for general/plains deployment; cost not optimised for altitude-specific derating losses.	Higher component-grade cost due to industrial and military grade parts, offset by extended operational life and reduced field-failure/maintenance cost at altitude.
Sensor Suite	Phased-array radar, RF sensors, EO/IR trackers, day/night camera, laser ranging.	mm Wave FMCW radar, EO camera, LWIR thermal (FLIR Lepton 3.5), passive RF sensing array, plus a dedicated secondary Environmental & System Health sensing layer (pressure, humidity, vibration, battery health).
High-Altitude Adaptability	Not specifically documented as altitude-optimised; assumes passive ruggedisation for nominal environmental conditions.	Purpose built: active PID gain scheduling, torque-residual gimbal compensation, zoned thermal management for low pressure cooling failure, cold induced voltage- ag mitigation.
Jamming / Interception Methods	RF communication link jamming, Anti GNSS/GPS spoofing (soft kill), Laser Directed Energy Weapon (hard kill).	Passive RF sensing for early warning; architecture is focused on detection, tracking and confidence-scored classification. Kinetic/DEW hard-kill layer is outside current scope; treated as a future integration point.
SWaP-C (Size, Weight, Power, Cost)	Static tripod-mounted or vehicle-mounted configuration; designed for general field deployment.	Compact IP65 NEMA enclosed node on tripod mount; LiFePO4 power with dedicated PDB and Smart BMS; power budget explicitly allocated for active thermal management, trading some power/weight margin for altitude specific reliability.